GitHub Mastery: Becoming a Pro Developer

A Concise Terminal Reference & Best Practices Guide | Vidhata Plastics Engineering

1. The Essential Daily Workflow

Command	Description / Action
`git status`	Check modified files, untracked files, and current branch state.
`git add .`	Stage ALL modified & new files for the next commit snapshot.
`git add <file>`	Stage a specific target file (e.g., git add css/style.css).
`git commit -m "msg"`	Save a staged checkpoint with a clear, descriptive commit message.
`git push <remote> <branch>`	Upload local commits to cloud (e.g. git push origin main).

2. Remotes & Synchronization

Command	Description / Action
`git remote -v`	List all connected remote repository URLs (fetch & push).
`git remote add upstream <url>`	Link a secondary remote repository (e.g., team/main repo).
`git fetch <remote>`	Download remote updates WITHOUT modifying local working code.
`git pull <remote> <branch>`	Fetch AND merge remote updates directly into active branch.
`git push upstream main`	Push local main branch commits to the upstream repository.

3. Professional Branching & Merging

Command	Description / Action
`git branch`	List local branches (* indicates currently active branch).
`git checkout -b feature-name`	Create AND switch to a new isolated feature branch.
`git checkout main`	Switch active context back to the main development branch.
`git merge feature-name`	Combine completed feature branch code into your current branch.
`git branch -d feature-name`	Safely delete a feature branch after successful merge.

4. Inspection, History & Safety Controls

Command	Description / Action
`git log --oneline -n 5`	Display a clean 1-line summary of the 5 most recent commits.
`git diff`	Review exact line-by-line code changes before staging.
`git restore <file>`	Discard uncommitted local edits (revert file to last commit).
`git restore --staged <file>`	Unstage a file without losing its modified code.
`git reset --soft HEAD~1`	Undo the last commit while keeping all code changes staged.

Pro Tip for Mastery: Always run git status before and after executing any staging or commit commands. It gives you 100% visibility over your code state!